Sahaj Bhadja

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Summary

Software engineer with 2+ years building production systems, automated testing workflows, and performance-sensitive applications. Strong in C++, Python, SQL, object-oriented design, CI/CD, and debugging complex software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected **multi-tenant** platform services on **Azure SQL** and Blob Storage for 500+ users, optimizing indexed queries and data-access patterns to reduce dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant background execution workflows with explicit state transitions and failure isolation, processing 1K+ AI jobs while improving throughput 3x under concurrent load and simplifying production debugging for releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Engineered **Flask** services and **REST APIs** for AI grading workflows serving 10K+ daily requests, integrating secure file ingestion, scoring pipelines, and instructor-facing features across customer-facing production software.
- Designed concurrent batch-processing with multithreading on **AWS RDS**, handling 1K+ file batches while maintaining stable API behavior, supporting release quality, and reducing turnaround time by 60% during peak use.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built spatial-temporal planning pipelines over 100K+ trajectory logs, integrating map geometry and agent context to improve closed-loop decision stability by 15% in safety-critical autonomous-driving simulations with tight feedback loops.
- Developed experiment and evaluation workflows for simulation-heavy systems, strengthening reproducibility, debugging, and failure analysis while collaborating on latency-sensitive model behavior across sensor-driven environments and repeated test cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **microservice**-based data-sharing software with **Jenkins**, **PyTest**, and **Mockito** automation across Windows, Linux, and macOS, improving release reliability through 20+ unit and integration test suites.
- Optimized large-scale **API** and **ETL** workflows with **AWS Glue**, **S3**, and object-oriented design patterns, improving response times by 30% and increasing pipeline efficiency by 68% in production.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
Systems & Engineering: object-oriented design, STL familiarity, algorithms, data structures, multithreading, debugging, performance tuning, unit testing, integration testing
Platforms & Tools: Git, Jenkins, Docker, Linux, Windows familiarity, AWS, Azure, GCP, Visual Studio familiarity, CI/CD
Backend & Data: REST APIs, Flask, FastAPI, Node.js, PostgreSQL, MySQL, SQL Server, Azure SQL, AWS RDS, S3, Blob Storage, ETL pipelines, microservices

Projects

CPU-scheduling Simulator Live | *C++, React, EJ-2*

- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with **C++** logic, dynamic metrics, and Gantt visualization across 1K+ simulated process combinations for repeatable systems experimentation and latency reasoning.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management while working across a large Java and TypeScript codebase with Docker-based local workflows.

FinNexus Live | *TypeScript, Kubernetes, MongoDB, Plaid*

- Built a scalable backend supporting 50,000+ users and 1M+ transactions, combining real-time synchronization, secure token workflows, and deployment automation to maintain reliable high-volume APIs under continuous activity.